

Project Report – Placement Portal System

1. Student Details

- **Name:** Ayendrila Sutradhar
- **Roll Number:** 23f2004330
- **Email:** 23f2004330@ds.study.iitm.ac.in
- **About Me:**
I am an undergraduate student pursuing **BS in Data Science from IIT Madras**. I am interested in **backend development, data science, and building scalable web applications**. I enjoy developing systems that automate manual processes and improve efficiency in real-world scenarios.

2. Project Details

Project Title

Placement Portal Management System

Problem Statement

Many colleges still manage placement activities using spreadsheets and manual communication. This often leads to issues such as:

- Difficulty tracking student eligibility
- Confusion about company recruitment schedules
- Lack of centralized student profiles
- Inefficient communication between students and administrators

To solve these issues, I developed a **Placement Portal System** that centralizes placement-related activities such as **student registration, company listings, and application tracking**.

Approach

The system was built using **Flask (Python)** as the backend framework and **SQLite** as the database. The application supports **multiple user roles** including:

- **Admin** – manages companies, job postings, and student records
- **Students** – view job opportunities and apply for positions

The system implements **role-based authentication** to ensure users can only access features relevant to them. The platform also includes dashboards for tracking applications and managing placement data efficiently.

3. AI / LLM Declaration

I used AI tools to assist with:

- Initial Flask project setup
- HTML template generation
- Debugging database model relationships
- Improving UI design with Bootstrap

Extent of AI Use:

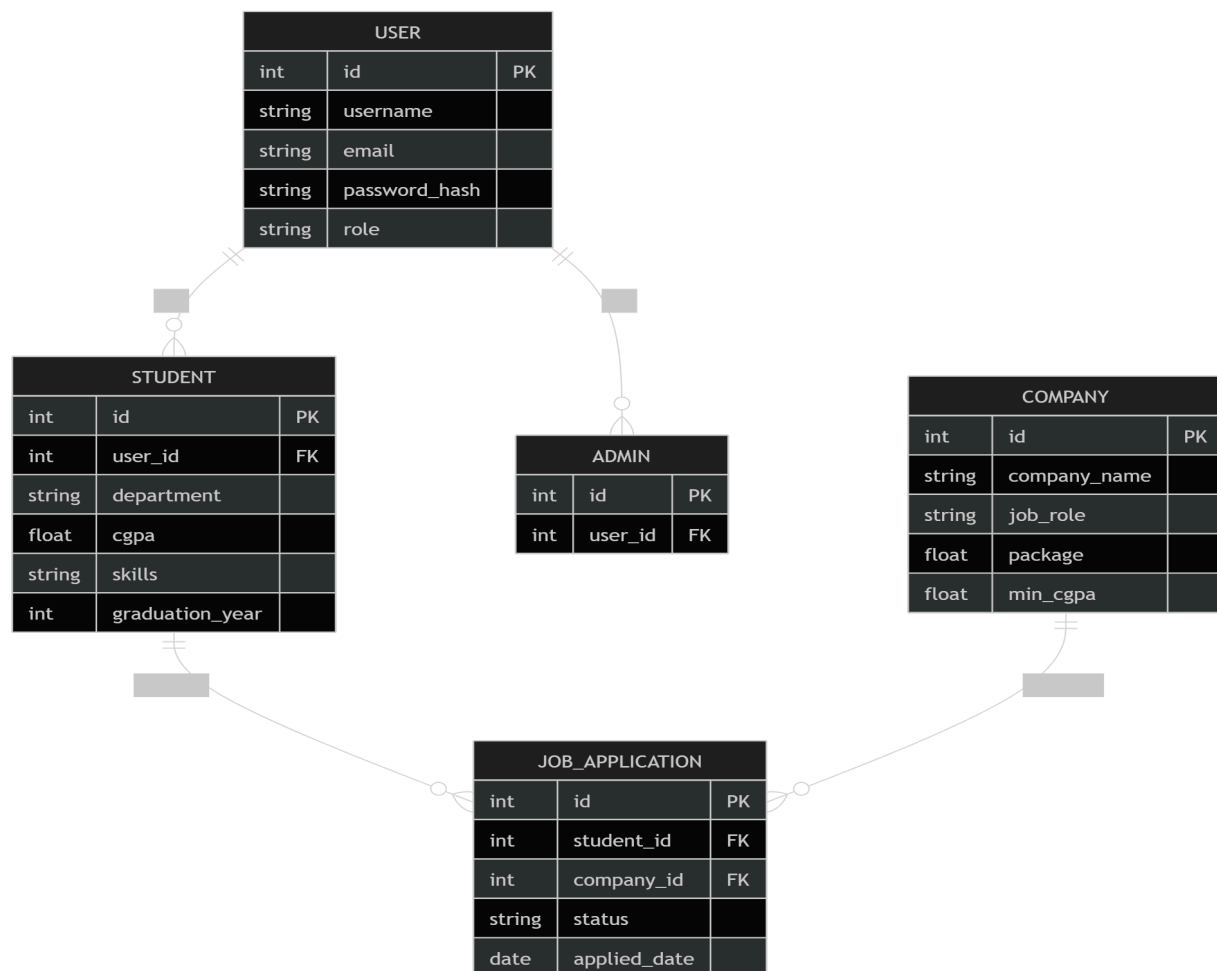
Approximately **25–30% of the code** was generated with AI assistance, mainly for **frontend styling and boilerplate structure**.

4. Technologies Used

Technology	Purpose
Flask	Backend framework for application logic

SQLite	Database to store students, companies, and applications
SQLAlchemy	ORM for database interaction
HTML	Frontend structure
Jinja2	Dynamic content rendering
Bootstrap 5	Responsive user interface
Flask-Login	Authentication and session management

5. Database Design (ER Diagram)



6. Application Routes (Endpoints)

Route	Method	Role	Description
/login	GET/POST	All	User login authentication
/register	GET/POST	Public	Student registration
/admin/dashboard	GET	Admin	Manage companies and students
/admin/add_company	GET/POST	Admin	Add new companies
/admin/view_applications	GET	Admin	View all student applications
/student/dashboard	GET	Student	View available companies
/student/apply/<company_id>	POST	Student	Apply for a job
/profile/edit	GET/POST	All	Update profile information
/logout	GET	All	End user session

7. Video Presentation

 report video.mp4